



Supporting Information

Optimizing Electrocatalytic Nitrogen Reduction via Interfacial Electric Field Modulation: Elevating d-Band Center in WS₂-WO₃ for Enhanced Intermediate Adsorption

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Experimental Procedures

Chemicals and Materials

Commercial WS₂ was obtained from Shanghai Chemical Reagent Co. Ltd. Sodium hydroxide (NaOH), hydrazine hydrate (N₂H₄·H₂O), and ethanol (C₂H₅OH) were procured from Beijing Tongguang Fine Chemical Co., Ltd. Hydrochloric acid (HCl), salicylic acid (C₇H₆O₃, ≥99.7%), ammonium chloride (NH₄Cl, 99.998%), p-dimethylaminobenzaldehyde (p-C₉H₁₁NO, 99%), trisodium sodium citrate dehydrate (C₆H₅Na₃O₇·2H₂O, 99%), sodium nitroferricyanide dihydrate (C₅FeN₆Na₂O·2H₂O, 99%), and sodium hypochlorite solution (NaClO, available chlorine 6–10%) were obtained from Beijing Chemical Corp. Nafion (5 wt%) was purchased from Sigma-Aldrich Chemical Reagent Co., Ltd. All chemicals were used without any further purification. Ultrapure Smart-S15 water (18.2 MΩ) was used in all experiments.

Preparation of Electrocatalysts

In brief, the commercial WS₂ was calcined in an air atmosphere at 500 °C for 2 hours to prepare WS₂-WO₃. WO₃ was prepared using the same method but was calcined at 800 °C for 2 hours in an air atmosphere.

Characterizations

To observe the morphology and microstructure of the samples, field emission scanning electron microscopy (SEM, FEI Quanta 200), transmission electron microscopy (TEM, FEI Tecnai G2 20), and high-resolution TEM (HRTEM, JEM-2100F, 200 kV) were employed. High-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) images were recorded on a JEOL JEM-ARM200F TEM/STEM with a spherical aberration corrector working at 300 kV. The crystalline structures were analyzed using X-ray diffraction (XRD, Rigaku D/max 2500) equipped with Cu Kα radiation (λ = 0.15405 nm). Ultraviolet photoelectron spectroscopy (UPS) measurements were conducted on Thermo Scientific ESCALab 250Xi using a gas discharge lamp with helium gas and the HeI (21.22 eV) emission line was employed. The work function values were calculated using the formula: $WF_{Exp} = 21.22 - (E_{cutoff} - E_{Fermi})$. The surface chemical status and composition were analyzed using X-ray photoelectron spectroscopy (XPS, Thermo Fisher Scientific) with an Al Kα X-ray source and FINDER Vista Laser micro-Raman Spectroscopy (Zolix, China) with laser excitation of 532 nm. The Kelvin probe force microscopy (KPFM) was obtained to analyze the surface potential of the materials using atomic force microscopy (AFM, Oxford MFP-3D, UK). For in situ Raman tests using an excitation wavelength of 532 nm, the electrochemical operando flow cell (EC-RAIR-H) was provided by Tianjin Gaoss Union technology Co. Ltd. A SOPTOP LMPlan objective (50X) was used for signal collection and the collection time was 30 seconds. Laser power was kept below 1.5 mW to protect the sample from laser damage.

Electrochemical Measurements

WS₂-WO₃/CP was prepared as follows: First, 5 mg of WS₂-WO₃ catalyst and 20 μL of Nafion solution (5 wt%) were dispersed in 980 μL of a mixed solution containing ethanol and H₂O (V:V=2:1) by sonication for 30 minutes to form a homogeneous ink. Then, 20 μL of the WS₂-WO₃ ink was loaded onto a carbon paper (1 cm × 1 cm) and dried. WS₂/CP and WO₃/CP were prepared using the same method. The Nafion membrane was pretreated by boiled in H₂O₂ solution (3 wt%), 0.5 M H₂SO₄, and ultrapure water at 80 °C for 1 h, respectively. Electrochemical measurements were carried out in a typical H-type electrolytic cell separated by Nafion 117 membrane. The prepared WS₂-WO₃/CP electrode, platinum gauze electrode and Ag/AgCl electrode were used as the working electrode, counter electrode and reference electrode, respectively. The potentials reported in this work were converted to RHE scale via calibration with the following equation: $E \text{ (vs RHE)} = E \text{ (vs Ag/AgCl)} + 0.197 + 0.0591 \times \text{pH}$. Before the electrochemical N₂ reduction measurements, N₂ was continuously pumped into the 0.1 M Li₂SO₄ electrolyte for 30 min. Electrochemical N₂ reduction was performed in N₂-saturated 0.1 M Li₂SO₄ electrolyte. The LSV curves were conducted at the scan rate of 10 mV s⁻¹ under Ar- or N₂-saturated 0.1 M Li₂SO₄ solution using the CHI 760e electrochemical workstation.

Determination of NH₃

Concentration of produced NH₃ was determined by the indophenol blue method. In detail, 2 ml post-tested solution was removed from the cathodic chamber, and then added into 2 ml 1.0 M NaOH solution containing C₇H₆O₃ and C₆H₅Na₃O₇·2H₂O (5 wt%). Next, 1 ml NaClO (0.05 M) and 0.2 ml Na₂[Fe(NO)(CN)₅]·2H₂O (1 wt%) aqueous solutions were added sequentially. The mixture was allowed to stand at room temperature for 2 hours, after which the UV-Vis absorption spectrum was measured. Calibration of the concentration-

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absorbance curve was performed using standard NH_4Cl solutions with concentrations ranging from 0.1 to 1.2 $\mu\text{g mL}^{-1}$, which were prepared by diluting a 100 $\mu\text{g mL}^{-1}$ NH_4Cl solution with 0.1 M Li_2SO_4 . The concentration of indophenol blue was determined by measuring the absorbance at a wavelength of 655 nm. A linear relationship between absorbance value and NH_4Cl concentration was observed with a fitting curve of $y = 0.2744x + 0.0339$ ($R^2 = 0.999$), which was obtained from three independent calibrations.

 ^{15}N isotope labeling experiment

$^{15}\text{N}_2$ with a ^{15}N enrichment of 99% (from Aladdin) was used to carry out the isotopic labeling experiment in a 0.1 M Li_2SO_4 electrolyte. The reactor was encapsulated, degassed, and injected with Ar for several times, and finally filled with $^{15}\text{N}_2$. After electrolysis at -0.3 V vs RHE for 2 h with the injection of 20 mL $^{15}\text{N}_2$ every 10 min, 30 mL of the electrolyte was taken out and concentrated to 1 mL by heating at 75 °C. Then, 0.1 mL of the resultant solution was thoroughly mixed with 0.6 mL of the dimethyl sulphoxide-D6 (99.9% D, from Adamas) for the ^1H nuclear magnetic resonance (NMR) measurement on a Bruker Avance III HD 500 NMR spectrometer. For comparison, $^{14}\text{N}_2$ experiment was also operated in the same method.

Determination of N_2H_4

The hydrazine present in the electrolyte was estimated via the method of Watt and Chrisp. A mixture of p- $\text{C}_9\text{H}_{11}\text{NO}$ (5.99 g), HCl (concentrated, 30 mL) and $\text{C}_2\text{H}_5\text{OH}$ (300 mL) was used as a color reagent. In detail, 2 mL electrolyte was removed from the electrochemical reaction vessel and added into 2 mL above prepared color reagent and stirring 20 min at room temperature. The absorbance of the resulting solution was measured at 455 nm. The concentration absorbance curves were calibrated using standard $\text{N}_2\text{H}_4 \cdot \text{H}_2\text{O}$ solution with a series of concentrations. The fitting curve ($y = 1.287x + 0.0109$, $R^2 = 0.9998$) shows a good linear relation of absorbance value with N_2H_4 concentration.

Calculations of NH_3 formation rate and Faradaic efficiency (FE)

The NH_3 yield rate was calculated using the following equation:

$$V(\mu\text{g h}^{-1} \text{mg}^{-1}) = \frac{C_{\text{NH}_3} \times V}{t \times m_{\text{cat}}}$$

The FE was calculated according to the following equation:

$$\text{FE} = \frac{3 \times F \times C_{\text{NH}_3} \times V}{17 \times Q} \times 100\%$$

where C_{NH_3} ($\mu\text{g mL}^{-1}$) is the measured concentration of NH_3 ; V (mL) is the volume of electrolyte (30 mL); t (s) is the reduction reaction time; m_{cat} (mg) is the mass loading of catalyst on CP; F is Faraday constant (96500 C mol^{-1}); Q (C) is the quantity of applied charge/electricity.

Computational Details

Spin-polarized density functional theory (DFT) simulations were performed to calculate the adsorption of N_2 molecule using the Vienna ab initio simulation package (VASP). The Perdew-Burke-Ernzerhof (PBE) functional within the framework of generalized gradient approximation (GGA) was applied to describe the exchange-correlation interactions. DFT-D2 method was adopted to consider the van der Waals (vdW) interaction. For structural optimization, the Brillouin zone was sampled by $4 \times 3 \times 1$ k-points for plane-wave basis, together with an energy cutoff of 500 eV. All the atoms were allowed to relax until the residue forces on each atom were less than 0.02 eV/Å. The convergence criterions were set to 10^{-5} eV for total energy. In order to avoid interactions between periodic images, a vacuum space of 25 Å was applied to all calculations. The formation Gibbs free energy (ΔG) of the NRR intermediates is calculated as:

$$\Delta G = \Delta E + \Delta \text{ZPE} - T\Delta S$$

where ΔE is the electronic energy difference directly obtained from DFT calculations, ΔZPE is the change of zero-point energies, T is the temperature ($T = 298.15 \text{ K}$), and ΔS is the entropy changes. The entropy of gas molecules is taken from standard values.

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Results and Discussion

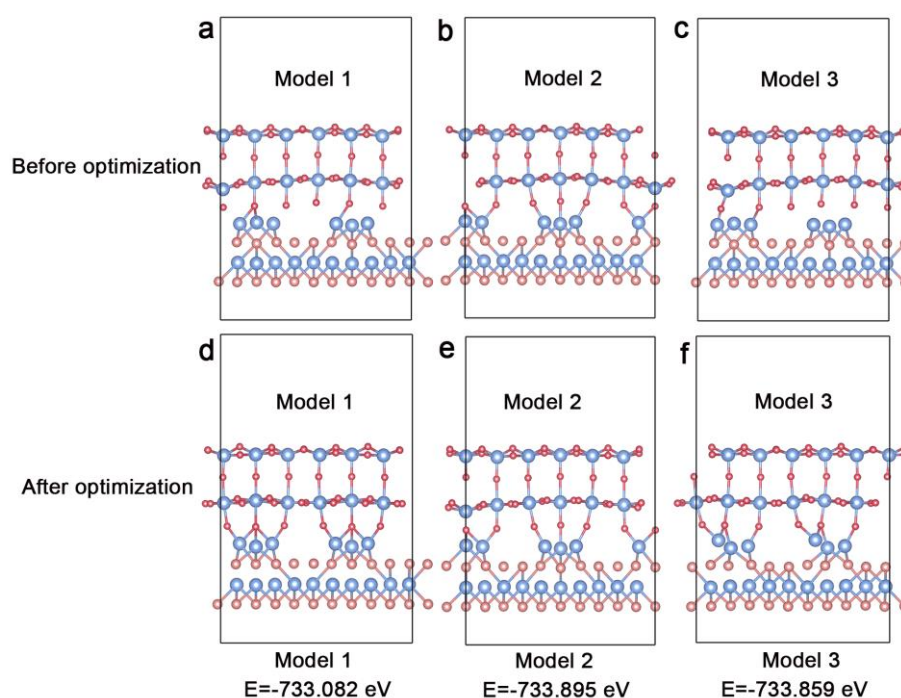


Figure S1. Three kinds of models from the side view before and after optimization for $\text{WS}_2\text{-WO}_3$.

We classify the interfaces between WS_2 and WO_3 into three cases before optimization. The first case is when the O atom is located directly above the triangular W atoms; the second case is when the triangular W atoms are all surrounded by four O atoms; and the last case is a middle case between the first two cases. To better distinguish them, we have defined them as Model 1, Model 2, and Model 3. It is evident that Model 2 is the most stable interface with the lowest energy, indicating that it is the most easily obtained structure experimentally.

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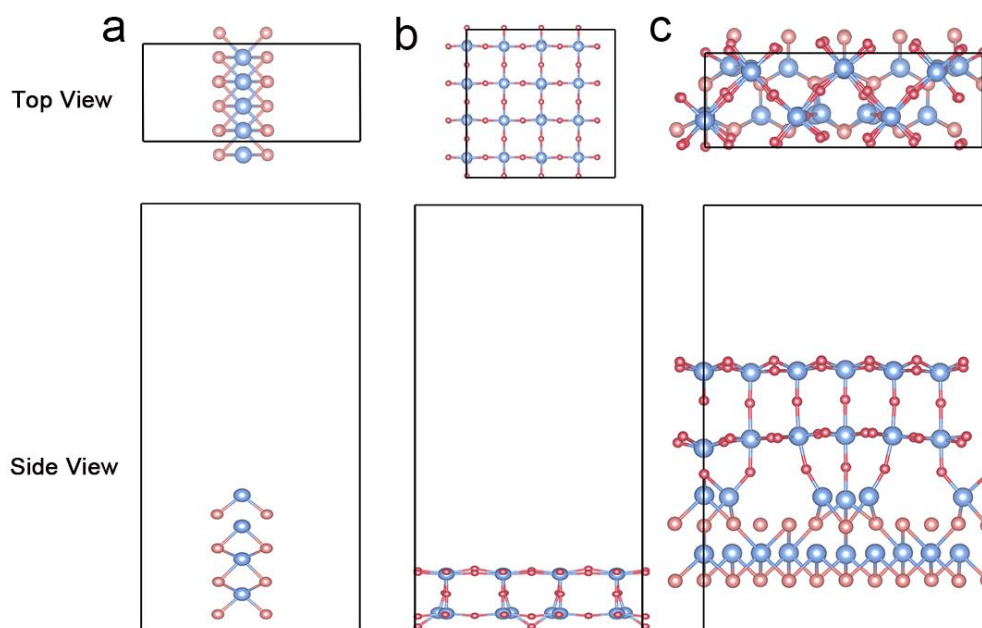


Figure S2. The top view (above) and side view (below) of optimized structure for (a) WS₂, (b) WO₃ and (c) WS₂-WO₃.

Table S1. The electrostatic potential difference between the top and bottom of WS₂-WO₃.

External electric field	Electrostatic potential difference
0 V/Angstrom	0.145 eV
0.1 V/Angstrom	0.174 eV
0.2 V/Angstrom	0.218 eV
0.3 V/Angstrom	0.286 eV
0.4 V/Angstrom	0.331 eV
0.5 V/Angstrom	0.355 eV

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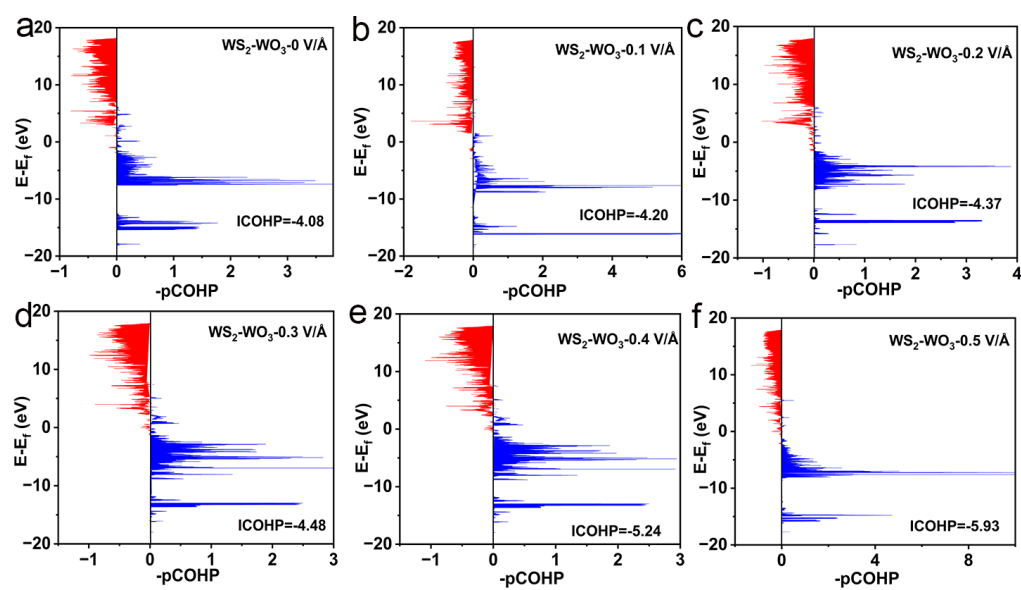


Figure S3. The pCOHP of $\text{WS}_2\text{-WO}_3$ with different interfacial electric field intensity.

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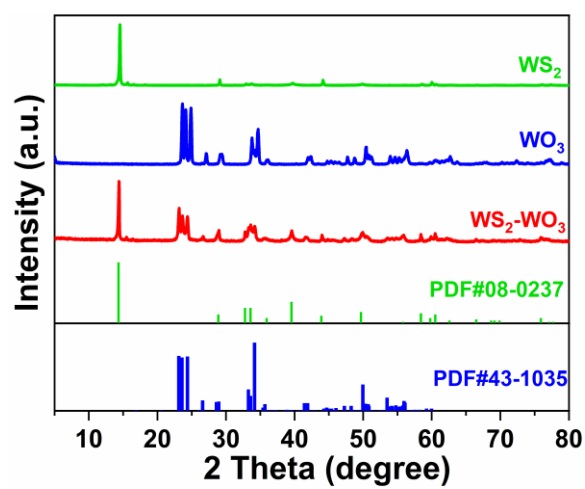


Figure S4. XRD patterns of WS₂, WO₃ and WS₂-WO₃.

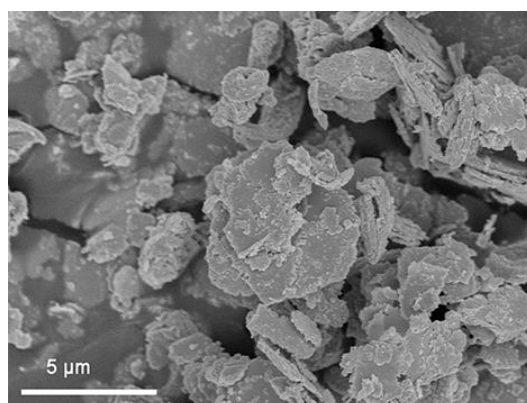


Figure S5. SEM image of WS₂-WO₃.

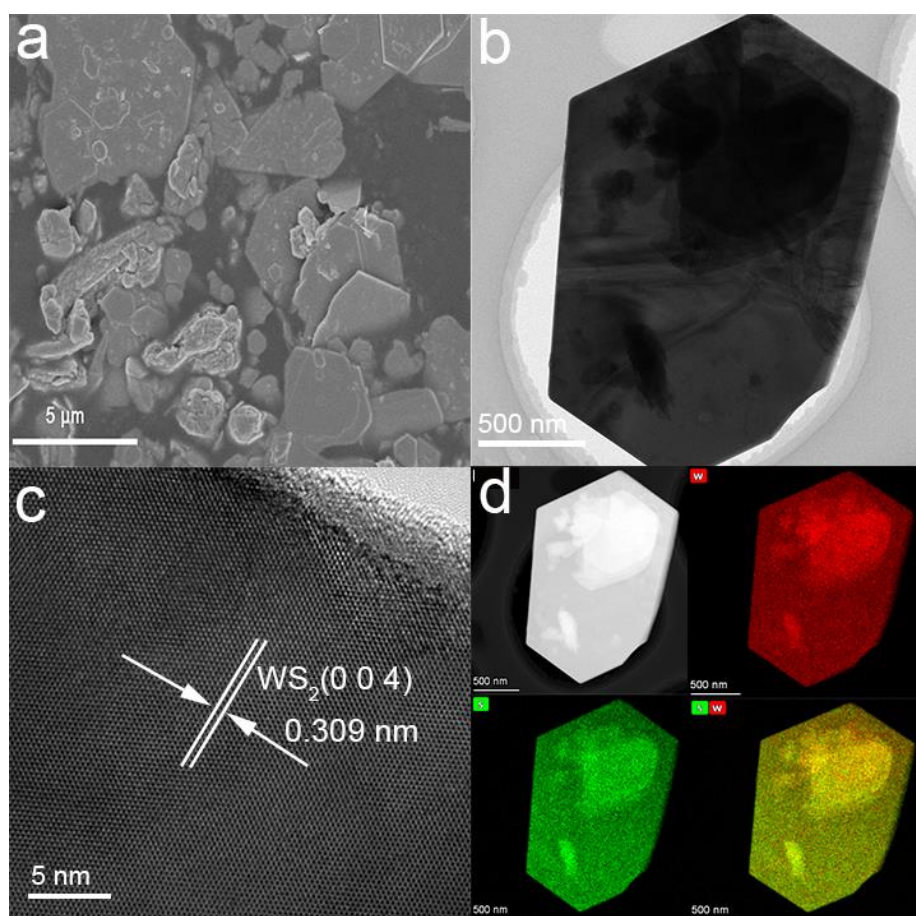


Figure S6. (a) SEM image of WS₂. (b) TEM image of WS₂. (c) HRTEM image of WS₂. (d) EDS images of WS₂.

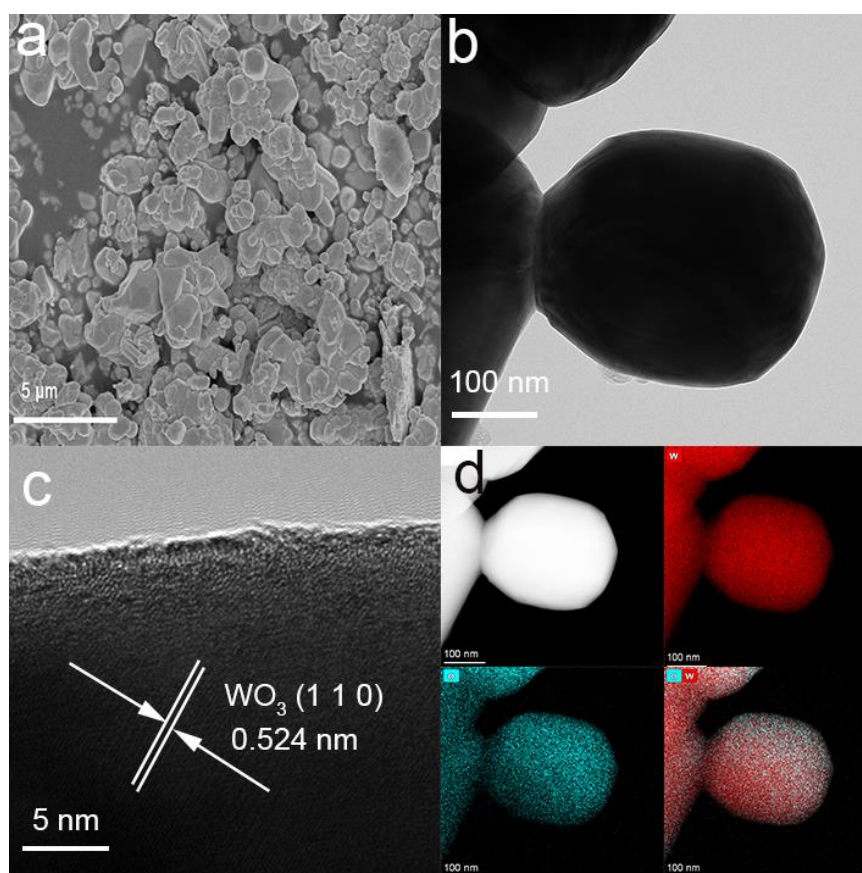


Figure S7. (a) SEM image of WO_3 . (b) TEM image of WO_3 . (c) HRTEM image of WO_3 . (d) EDS images of WO_3 .

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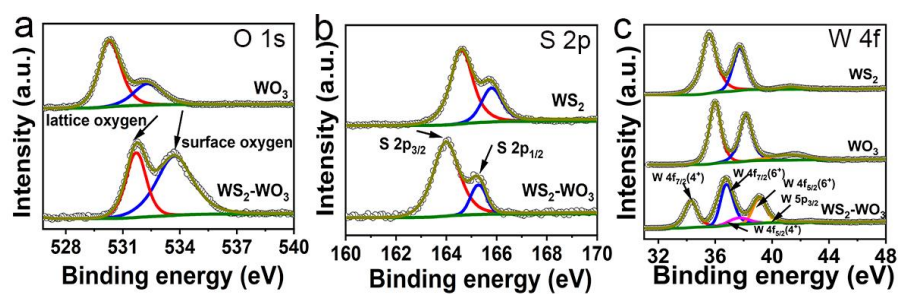


Figure S8. XPS spectra of (a) O 1s for WO_3 and $\text{WS}_2\text{-WO}_3$. (b) S 2p for WS_2 and $\text{WS}_2\text{-WO}_3$ and (c) W 4f for WS_2 , WO_3 and $\text{WS}_2\text{-WO}_3$.

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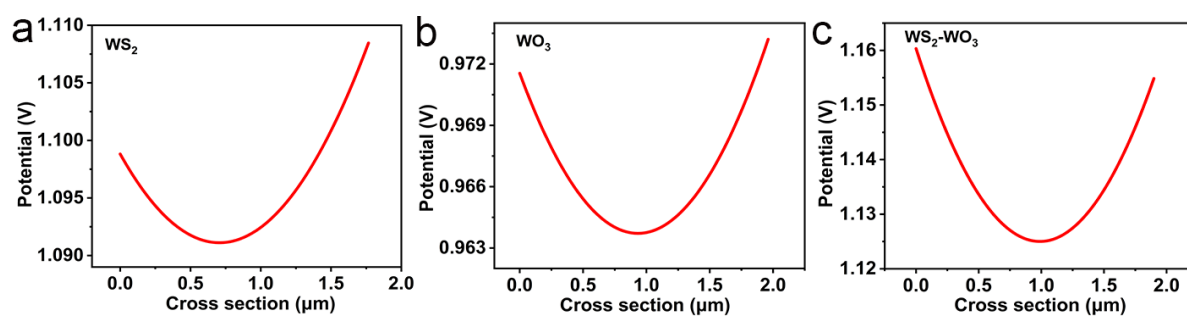


Figure S9. The potential distributions of interfacial electric field on (a) WS_2 , (b) WO_3 and (c) $\text{WS}_2\text{-WO}_3$.

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Figure S10. The home-made reactor for ENRR.

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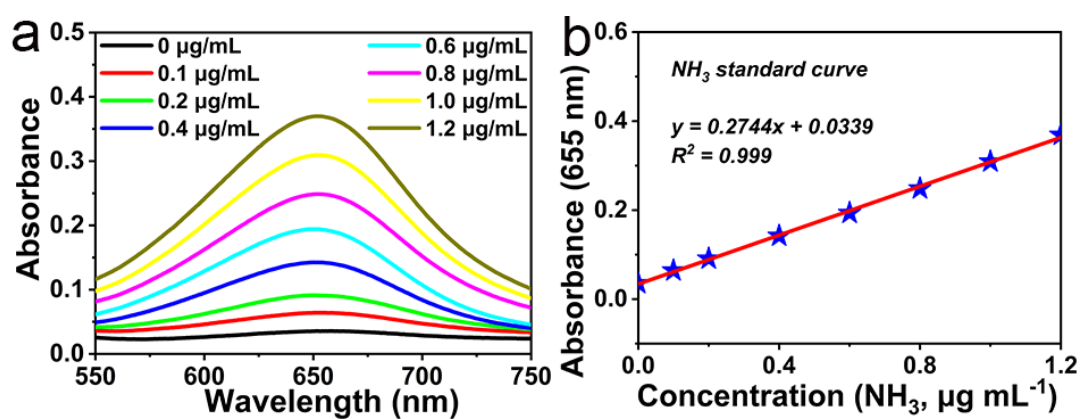


Figure S11. (a) UV-Vis absorption spectra of various NH_4^+ concentrations after incubated for 2h at room temperature. (b) Calibration curve used for calculating NH_4^+ concentrations.

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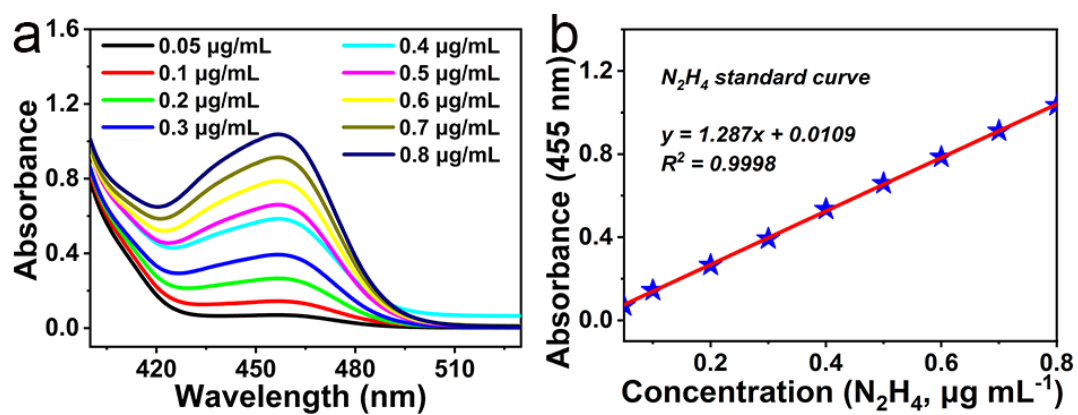


Figure S12. (a) UV-Vis absorption spectra of various N_2H_4 concentrations after incubated for 20 min at room temperature. (b) Calibration curve used for calculation of N_2H_4 concentrations.

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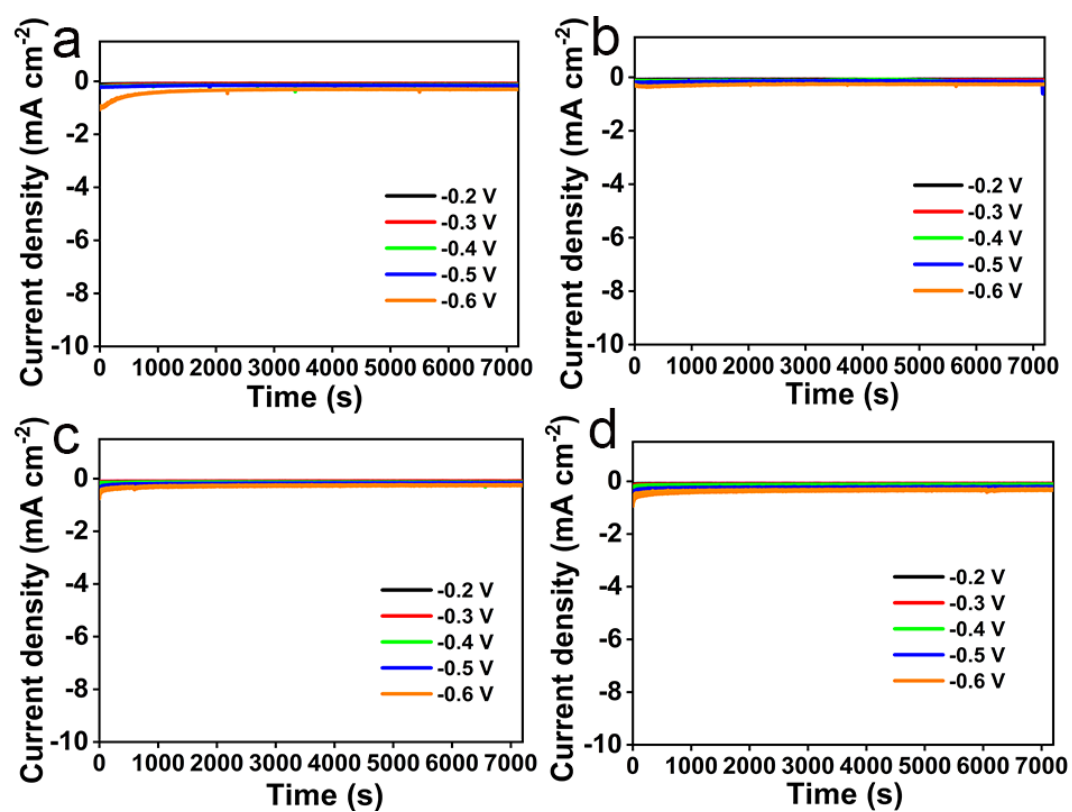


Figure S13. Chronoamperometry curves of (a) WS_2 , (b) WO_3 , (c) $\text{WS}_2\&\text{WO}_3$ and (d) $\text{WS}_2\text{-WO}_3$ at different potentials in N_2 -saturated $0.1 \text{ M Li}_2\text{SO}_4$.

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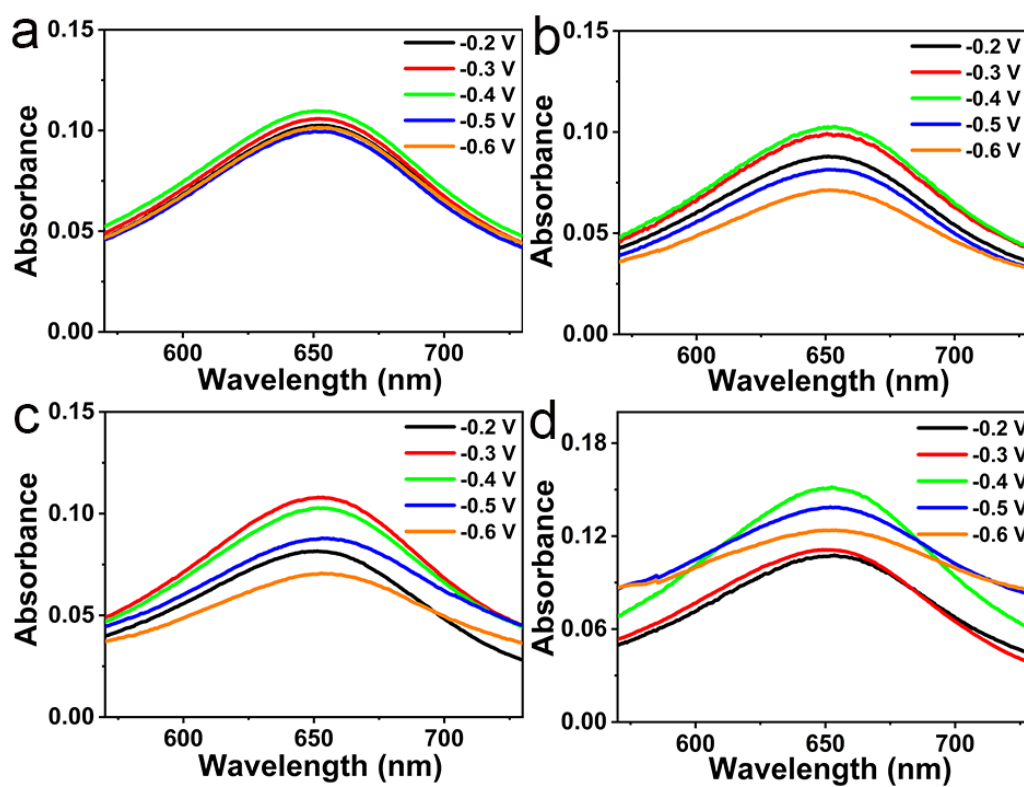


Figure S14. UV-Vis absorption spectra of (a) WS₂, (b) WO₃, (c) WS₂&WO₃ and (d) WS₂-WO₃, which records absorbance of different potentials.

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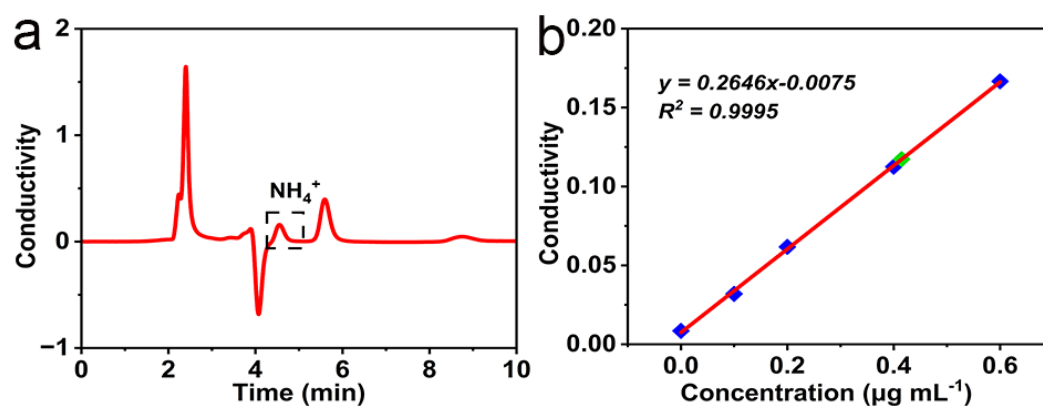


Figure S15. (a) Ion chromatogram for the NH_4^+ ions. (b) Calibration curve used for the estimation of NH_4^+ .

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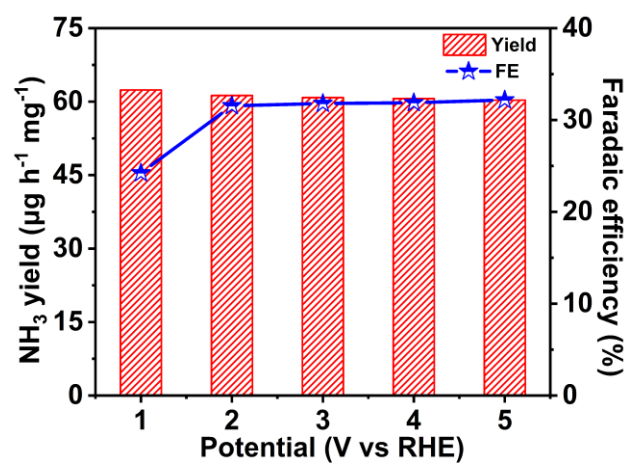


Figure S16. NH₃ yields and FEs of WS₂-WO₃ at -0.4 V vs RHE during 5 recycling tests.

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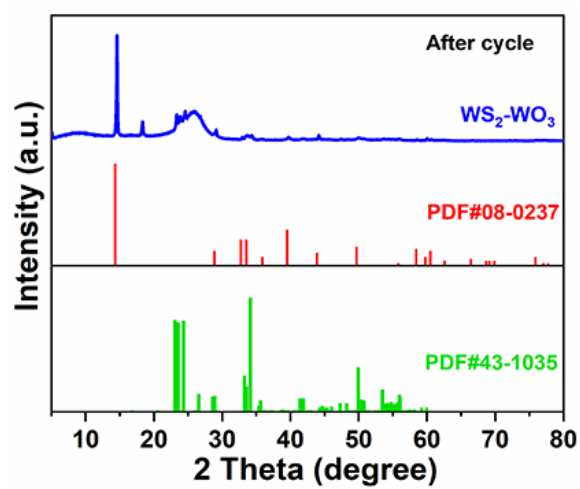


Figure S17. XRD patterns of WS₂-WO₃ after 5 recycling tests.

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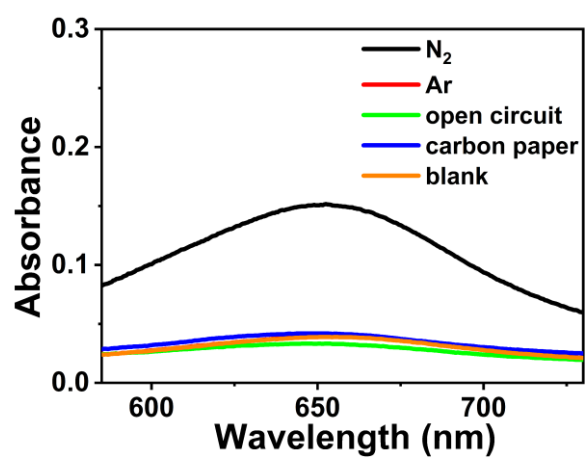


Figure S18. UV-vis absorption spectra at different control conditions.

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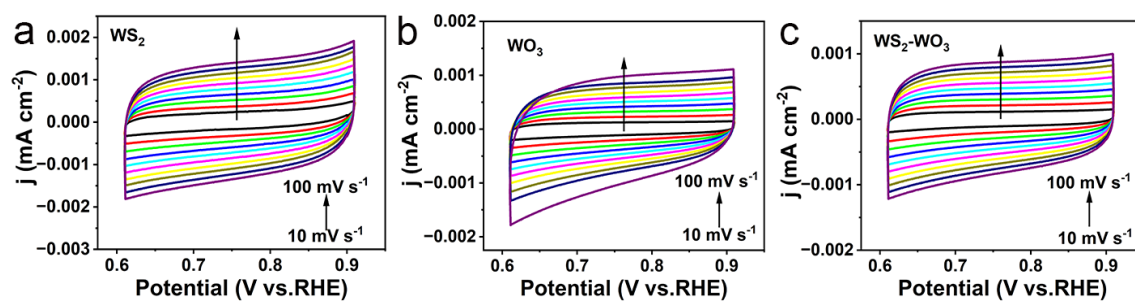


Figure S19. CV curves of (a) WS_2 , (b) WO_3 and (c) $\text{WS}_2\text{-WO}_3$.

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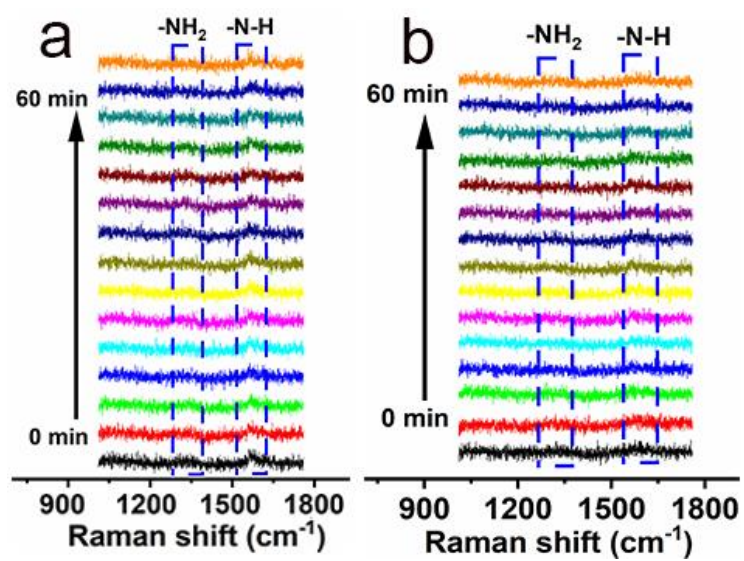


Figure S20. In-situ Raman characterization of (a) WS₂ and (b) WO₃ at -0.4 V in 0.1 M Li₂SO₄.

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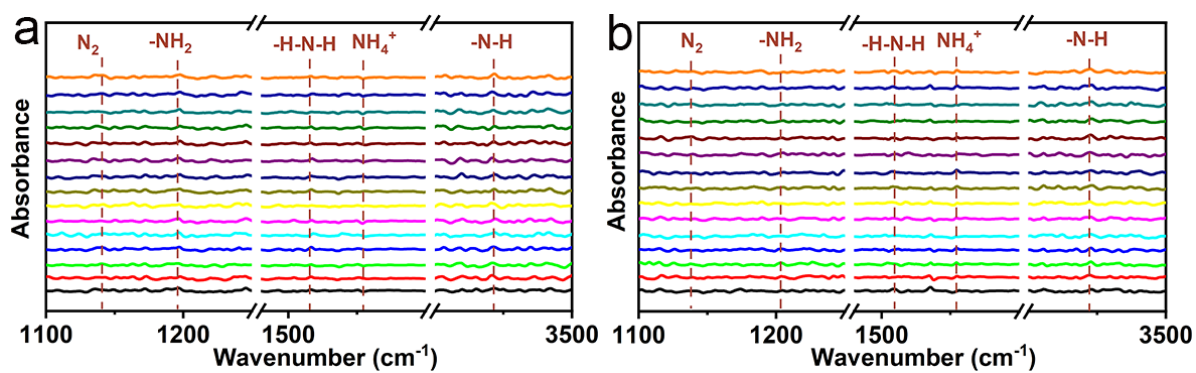


Figure S21. In-situ FTIR spectra of (a) WS₂ and (b) WO₃ at -0.4 V in 0.1 M Li₂SO₄.

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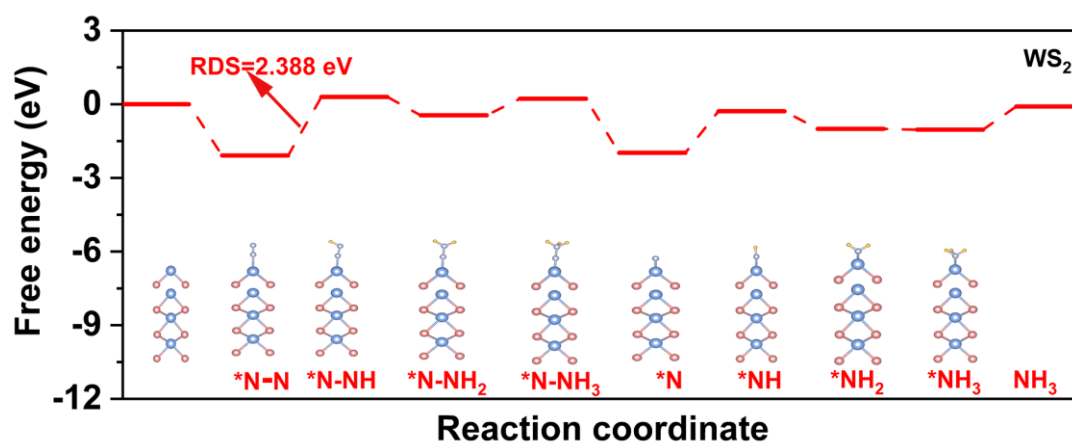


Figure S22. Free energy diagram for NRR of WS₂ with the optimized structures of different intermediates without incorporating electric field. This is due to the absence of an electrostatic potential difference between the top and bottom surfaces of WS₂.

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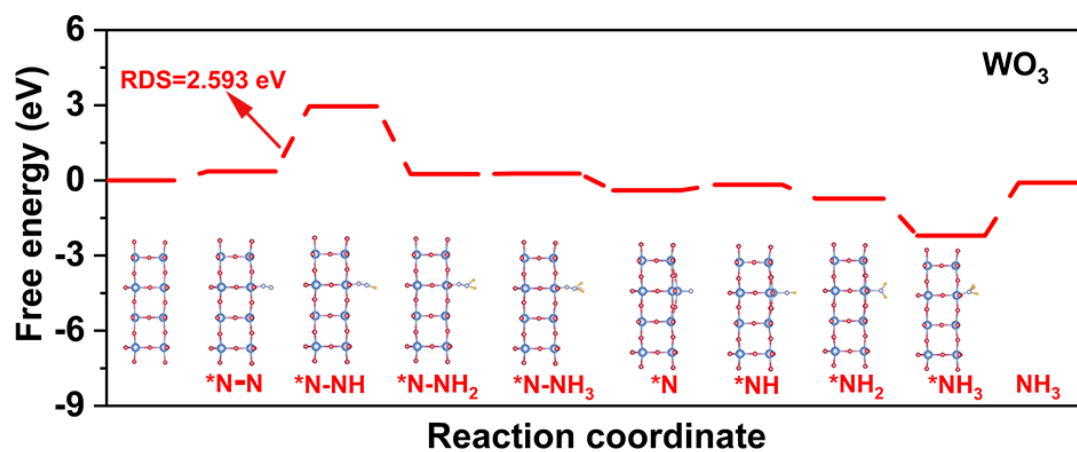


Figure S23. Free energy diagram for NRR of WO_3 with the optimized structures of different intermediates without incorporating electric field. This is due to the absence of an electrostatic potential difference between the top and bottom surfaces of WO_3 .

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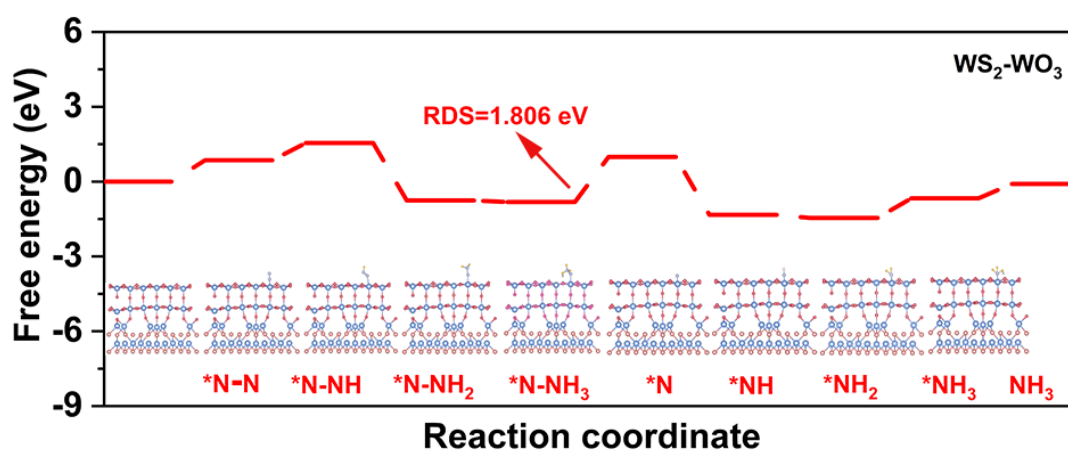


Figure S24. Free energy diagram for NRR of $\text{WS}_2\text{-WO}_3$ with the optimized structures of different intermediates considering its inherent interfacial electric field. This is due to the presence of a significant electrostatic potential difference between the top and bottom surfaces of the $\text{WS}_2\text{-WO}_3$ system.

Table S2. The charge transfer of $\text{WS}_2\text{-WO}_3$ during ENRR.

intermediate	charge transfer
*N-N	0.152312
*N-NH	0.069149
*N-NH ₂	0.283104
*N-NH ₃	0.331379
*N	0.070194
*NH	0.530834
*NH ₂	0.837389
*NH ₃	0.241738

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Table S3. Comparison of electrocatalytic N₂ reduction performance with other recently reported electrocatalysts.

Catalysts	Electrolyte	Yield (NH ₃)	FE (NH ₃)	Ref.
WS ₂ -WO ₃	0.1 M Li ₂ SO ₄	62.38 µg h ⁻¹ mg _{cat} ⁻¹	24.24%	This work
FeSA-NSC	0.1 M HCl	30.4 µg h ⁻¹ mg _{cat} ⁻¹	21.9%	[1]
Zn ¹ N-C	0.1 M KOH	16.1 µg h ⁻¹ mg _{cat} ⁻¹	11.8%	[2]
FePc-pz	0.01 M H ₂ SO ₄	33.6 µg h ⁻¹ mg _{cat} ⁻¹	31.9%	[3]
MoSAs-Mo ₂ C/NCNTs	0.005 M H ₂ SO ₄ :0.1 M K ₂ SO ₄	16.1 µg h ⁻¹ mg _{cat} ⁻¹	7.1%	[4]
MoO _{2+x}	0.1 M KOH	3.95 µg h ⁻¹ mg _{cat} ⁻¹	22.1%	[5]
CoP/CNT	0.1 M Na ₂ SO ₄	39.58 µg h ⁻¹ mg _{cat} ⁻¹	19.4%	[6]
Mo ₂ C-MoO ₂	0.1 M H ₂ SO ₄	13.94 µg h ⁻¹ mg _{cat} ⁻¹	12.72%	[7]
Au/CoO _x	0.5 M H ₂ SO ₄	15.1 µg h ⁻¹ cm ⁻²	19%	[8]
BiNi	0.1 M Na ₂ SO ₄	17.5 µg h ⁻¹ mg _{cat} ⁻¹	13.8%	[9]
W-NO/NC	0.5 M LiClO ₄	12.62 µg h ⁻¹ mg _{cat} ⁻¹	8.35%	[10]
FeSA-NO-C	0.1 M HCl	31.9 µg h ⁻¹ mg _{cat} ⁻¹	11.8%	[11]
VS ₂ -350	0.1 M HCl	20.29 µg h ⁻¹ mg _{cat} ⁻¹	3.86%	[12]
D-BP _{ene}	0.1 M PBS	70 µg h ⁻¹ mg _{cat} ⁻¹	26%	[13]
Co _x Ni _{3-x} (HITP) ₂ /BNSs-P	0.1 M HCl	128.26 µg h ⁻¹ mg _{cat} ⁻¹	52.92%	[14]
1T-MoS ₂	0.1 M Na ₂ SO ₄	71.01 µg h ⁻¹ mg _{cat} ⁻¹	21.01%	[15]

SUPPORTING INFORMATION

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